Name of Subject: Data Analytics Lab
Name of Student:
Roll No:
Class: SE E & TC
Division:
Academic Year: 2023-24

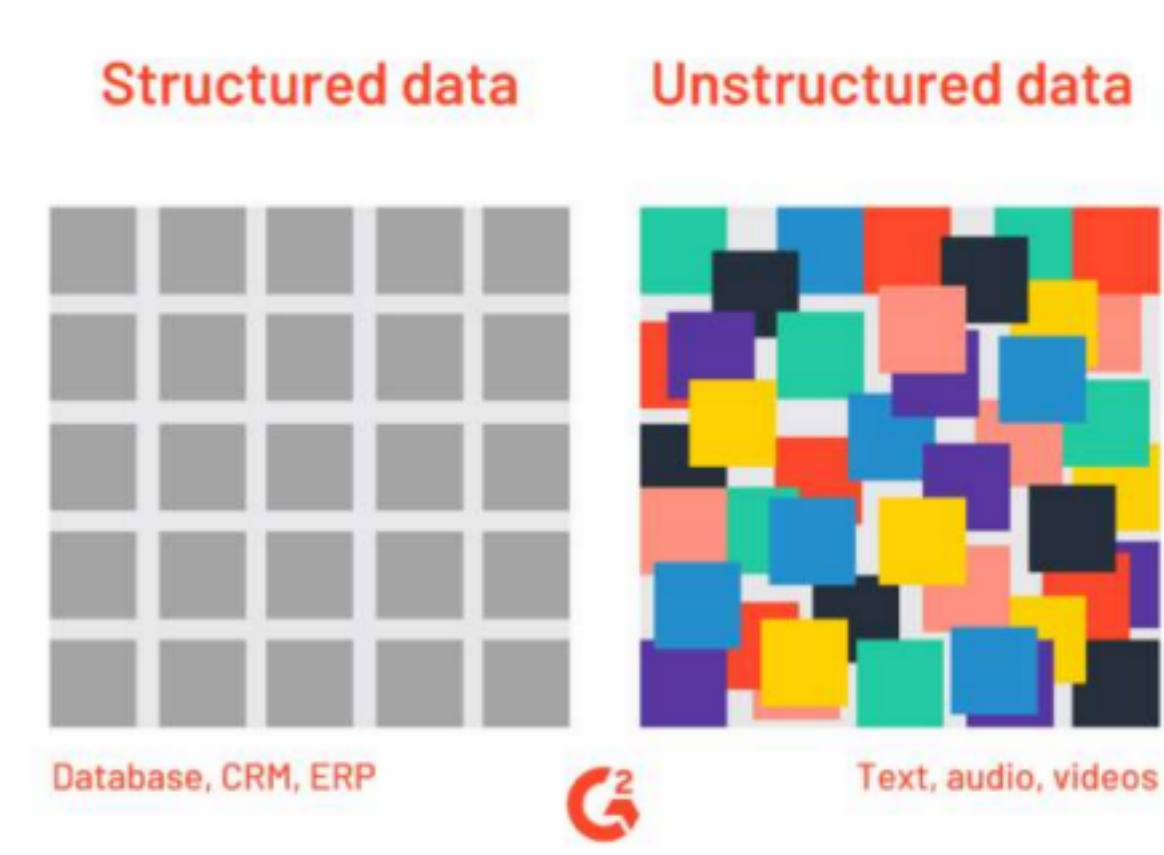
Assignment 1: Basic Programming in python

a. Perform Addition, subtraction, Multiplication for given numbers
b. Import String from the user and perform Sub operations on the same
c. Print a half and hollow pyramid pattern using a nested loop.
d. Find Even or Odd of given numbers
e. Find the largest number among three using function

Theory:

Data Science:

It is a field that is used to extract information from structured or unstructured data.



Data science is related to data mining, Machine Learning, Big data and is the field of study that combines

- o Domain expertise
- o Programming skills
- o Knowledge of mathematics and statistics.
- o To extract meaningful insights from data.

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Data Analytics:

Analytics is defined as "the scientific process of transforming data into insights for making better decisions"

Analytics, is the use of data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models to help managers gain improved insight about their business operations and make better, fact-based decisions – James Evans

Classification of Data analytics

There are four major types of data analytics.

• Descriptive analytics

• Diagnostic analytics

• Predictive analytics

• Prescriptive analytics Descriptive Analytics

Descriptive Analytics

• Descriptive Analytics, is the conventional form of data analysis

• It seeks to provide a depiction or "summary view" of facts and figures in

• an understandable format

• This either inform or prepare data for further analysis

• Descriptive analysis or statistics can summarize raw data and convert it into a form that can be easily understood by humans

They can describe in detail about an event that has occurred in the past A common example of Descriptive Analytics are company reports that simply provide a historic review like:

• Data Queries

• Reports

• Descriptive Statistics

• Data Visualization

• Data dashboard

Diagnostic analytics

• Diagnostic Analytics is a form of advanced analytics which examines data

• or content to answer the question "Why did it happen?"

• Diagnostic analytical tools aid an analyst to dig deeper into an issue so that they can arrive at the source of a problem

• In a structured business environment, tools for both descriptive and diagnostic analytics go parallel

It uses techniques such as:

• Data Discovery

• Data Mining

• Correlations

Predictive analytics

• Predictive analytics helps to forecast trends based on the current events

• Predicting the probability of an event happening in future or estimating

• the accurate time it will happen can all be determined with the help of •

predictive analytical models

• Many different but co-dependent variables are analysed to predict a trend in this type of analysis

Set of techniques that use model constructed from past data to predict the future or ascertain impact of one variable on another:

1. Linear regression

2. Time series analysis and forecasting

3. Data mining

Prescriptive analytics

• Set of techniques to indicate the best course of action

• It tells what decision to make to optimize the outcome

The goal of prescriptive analytics is to enable:

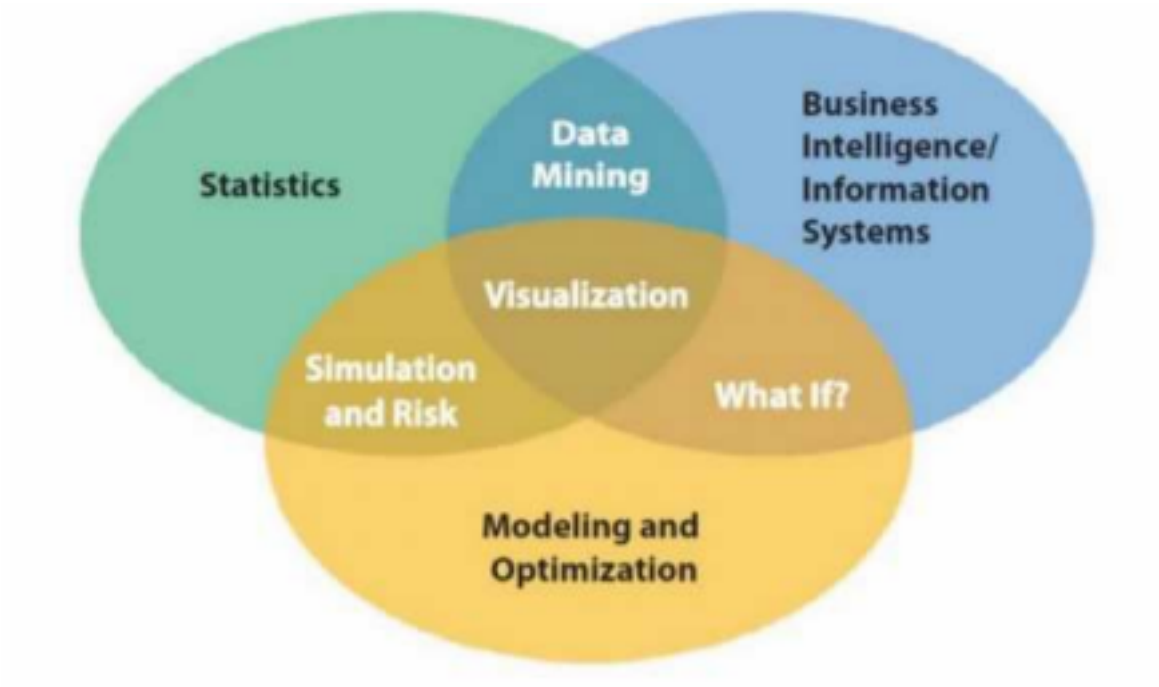
1. Quality improvements

2. Service enhancements

3. Cost reductions and

4. Increasing productivity

Elements of data Analytics



Python:

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. Can streamline large complex data sets.

It was created by Guido van Rossum during 1985- 1990.

Python is an easy to learn, powerful programming language

Extensive library support

graphical options along with visualization tools

Extended Pack of Analytics Tools Available

High readability, Easy to learn

Python Packages for Data Science

Python Basics

Variables are containers to store data.

Python has no command for declaring a variable.

A variable is created the moment you first assign a value to it.

Variables do not need to be declared with any particular type, and can even change type after they have been set.

Variable names are case-sensitive.

We can assign multiple values to a variable.

Global Variables

- o Variables that are created outside of a function (as in all of the examples above) are known as global variables.
- o Global variables can be used by everyone, both inside of functions and outside.

Python has the following data types built-in by default, in these categories:

Text Type: Str

Numeric Types: int, float, complex

Sequence Types: list, tuple, range

Mapping Type: Dict

Set Types: set, frozenset

Boolean Type: Bool

Binary Types: bytes, bytearray, memoryview

Strings in python

Python string is the collection of the characters surrounded by single quotes, double quotes, or triple quotes. The computer does not understand the characters; internally, it stores manipulated character as the combination of the 0's and 1's.

Each character is encoded in the ASCII or Unicode character. So we can say that Python strings are also called the collection of Unicode characters.

Operators

Arithmetic Operators, Comparison Operators, Logical Operators, and String Operators can be used for Python Programming

Control Structures

Loops in Python

- o for
- o while

Conditional statement

- o If - else

Lists vs Tuples

List	Tuple
Mutable	Unmutable
Not efficient in memory usage	Uses memory efficiently
Slower access	Faster access
Cannot be used a key to dictionary	Can be used as key to dictionary
Slicing possible	Slicing not possible

Can create an empty list	Cannot create a tuple of single element

Functions

- o Reusability
- o Readability
- o Debugging
- o Development time

Pass by value

Function definition is here

```
def changeme( mylist ):
mylist.append([1,2,3,4])
print "Values inside the function: ", mylist
return
# Now you can call changeme function
mylist = [10,20,30];
changeme( mylist );
print(mylist)
```

Conclusion:

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